Hamiltonian Vision Kernel: A Physics-Informed Hybrid Quantum–Classical Autoencoder with Provable Symmetry and Real Hardware Validation

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***Abstract*—We introduce the Hamiltonian Vision Kernel (HVK), a physics-informed framework for hybrid quantum–classical visual representation and reconstruction. HVK combines matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder. HVK1D and HVK2D instantiate chain and grid interaction graphs, while observable pooling provides D_4 equivariance by construction. In a scoped three-image-per-dataset study across six real image datasets, the framework supports high-fidelity per-image reconstruction. Three results characterize its structure. First, the pooled HVK2D map achieves numerical equivariance error of 9.57×10^{-17}. Second, across five seeds on a leakage-audited task requiring distant-patch products, the entangling observable channel reaches mean $R^2 = 0.974$, compared with $R^2 \leq 0.02$ for non-entangling controls. Third, five images are reconstructed from observables measured on IBM Quantum hardware without retraining, reaching 25.90–31.52 dB PSNR. These results motivate HVK as a framework for studying symmetry, topology, nonlocal correlations, Hamiltonian diagnostics, and hardware-executed visual models. A companion supplementary study provides resource-matched held-out controls, leakage audits, statistical controls, and complete hardware methodology.**

***Index Terms*—Quantum Image Processing, Tensor Networks, Variational Quantum Algorithms, Pauli Observables, Hamiltonian Diagnostics, Group-Equivariant Representations, Hybrid Quantum Hardware Validation.**

I. Introduction

Hybrid quantum–classical models for vision tasks are increasingly proposed as a route to representations with favorable inductive bias, exploiting entanglement, physically motivated regularizers, or tensor-network structure that classical convolutional pipelines do not natively expose [1], [2], [20]. The variational quantum eigensolver and its descendants exploit the local decomposability of physically relevant Hamiltonians to construct shallow, problem-aware circuits [1]–[3], and quantum image processing provides encodings that place pixel information directly into qubit registers [4]–[6]. Tensor-network methods, in parallel, have proven effective as compressed representations of structured classical data, including images [7]–[11], [22], and more broadly other structured data such as particle-physics events [16]. Related hybrid quantum architectures target adjacent imaging tasks, including quantum annealing-based tomographic reconstruction [12], adiabatic quantum computing for emission tomography [14], quantum neural compressive sensing for ghost imaging [13], hybrid autoencoder-plus-quantum-SVM feature extraction for image classification [17], and quantum capsule networks [18]. HVK focuses on the specific combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the Hamiltonian Vision Kernel (HVK1D/HVK2D), which combines these ingredients into a single architecture with three properties studied jointly here: a *Pauli-correlator latent space* (measured VQC observables, not a raw quantum state, form the representation handed to a classical decoder, in the spirit of quantum-enhanced feature maps [21]); an *enforced, provable D_4 symmetry* extension, group-averaging the observable grid over the eight symmetries of the square rather than merely motivating symmetry through architecture choices, in the spirit of group-equivariant classical vision [23] and symmetry-embedding constructions for variational circuits [24]; and a *learnable Hamiltonian energy diagnostic* (full nearest-neighbor $XX/YY/ZZ$ sectors in HVK1D and grid-edge ZZ terms in HVK2D). Table VII contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

a) Contributions.:

- **The HVK1D/HVK2D architecture**: MPS patch features, a 6-qubit VQC producing a 19–27-D Pauli-correlator observable vector, 2D Fourier positional encoding, a learnable graph-Hamiltonian energy diagnostic, and a classical decoder, with an optional D_4-pooled observable-grid variant.
- **Reconstruction results across six real image datasets** (CIFAR-10, MNIST, Fashion-MNIST, PathMNIST, BloodMNIST, PneumoniaMNIST), each trained per-image under an identical protocol (Section IV-A).
- **A provable D_4 symmetry extension**: a group-pooled HVK2D observable map is equivariant to the square dihedral group by construction, verified to floating-point precision (Section IV-E).
- **A restricted entanglement-sensitive result**: within a leakage-audited feature class and fixed linear-readout protocol, entangling pair observables are necessary when the target explicitly depends on distant-patch products (Section IV-F).
- **A real IBM-hardware reconstruction pilot** (Sec-

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tion IV-C): the exact parameters of an already-trained checkpoint are replayed on real quantum hardware with no retraining, and five images are reconstructed directly from hardware-measured observables.

A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, reports resource-matched held-out comparisons of HVK and classical/ablated feature maps, together with full hardware-pilot methodology (circuit construction, validation, job identifiers, quota accounting) and explicit provenance audits of excluded exploratory results. We keep that analysis in a companion document so the architecture-level contributions above and the generalization question do not obscure each other; Section V-B summarizes its central finding and how it relates to the results here.

II. Hamiltonian Vision Kernel: Architecture (Compact)

HVK treats each spatial patch as a finite spin state, compresses it as a matrix product state (MPS), measures a fixed set of Pauli observables from a variational quantum circuit (VQC), and decodes those observables back into pixels with a classical multi-layer perceptron (MLP). A learnable graph-Hamiltonian energy term regularizes training; unlike parameterized Hamiltonian learning approaches that treat the Hamiltonian as the primary learned object [19], HVK's energy term is a diagnostic regularizer alongside a separate reconstruction objective. The shared pipeline is:

$$\begin{aligned} \text{Image} &\to \{\text{Patches}\} \to \text{MPS} \\ &\to \text{46-D/22-D Feature} \to \text{6-D Vector} \\ &\to \text{27-D/19-D Latent} \to \text{Decoder} \\ &\to \text{Reconstruction.} \end{aligned} \tag{1}$$

Two executed preprocessing configurations share this pipeline. For the 256×256 *Monalisa*/HVK1D benchmark, sixteen non-overlapping 64×64 patches are ℓ_2-normalized, reshaped as 12-site spin states, and sequentially SVD-compressed to MPS bond dimension $\chi = 4$. Their 24 local X/Z expectations, 11 nearest-neighbor ZZ correlations, and 11 bipartite entropies form a 46-D feature vector. For the native 32×32 HVK2D multi-dataset experiments, sixteen 8×8 patches are represented by six-site MPSs, giving 12 local expectations, five nearest-neighbor ZZ terms, and five entropies (22 dimensions). A learned linear layer compresses either MPS feature vector to six circuit angles. HVK1D then measures 27 observables (six local Z, six local X, and five each of ZZ, XX, and YY); HVK2D measures 19 (six local Z, six local X, and seven grid-edge ZZ terms). The respective Fourier positional vectors have dimension eight and four. In both configurations, a two-hidden-layer MLP (128 and 256 units) maps the observable and positional vectors to one grayscale patch. Figure 1 shows the exact HVK1D and HVK2D ansatz circuits (rebuilt in Qiskit and numerically verified against the training-time PennyLane circuit as part of the hardware pilot of Section IV-C); the Hamiltonian energy-loss definitions and the D_4-pooling construction are given in Appendix A.

TABLE I
HVK variants tested in this paper.

Variant	Topology	Latent dim.	Symmetry
HVK1D	6-qubit chain	27-D	None (control)
HVK2D	2×3 grid	19-D	D_4-motivated only
Symmetric HVK1D	6-qubit chain	27-D	U(1) coupling
D_4-pooled HVK2D	2×3 grid, pooled	19-D	D_4-equivariant (proven)

Table I summarizes the tested variants. HVK1D uses a linear 6-qubit chain (5 nearest-neighbor bonds, 27-D observable vector); HVK2D arranges the same 6 qubits on a 2×3 lattice matching patch-boundary adjacency (19-D latent signature); a symmetric HVK1D variant adds a $\mathrm{U}(1)$-preserving coupling constraint; and a D_4-pooled HVK2D variant (Section IV-E) group-averages the observable grid over the eight symmetries of the square, $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g \in D_4} P_g^{-1}\Phi(gx)$, which is equivariant by construction. An IBM Cloud circuit probe (Appendix A) confirms both HVK1D and HVK2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

III. Experimental Setup

All image-reconstruction results in Sections IV-A–IV-C use the single-image *Monalisa* protocol developed for HVK: a fresh model is optimized directly on its target image, with no held-out split. They therefore measure per-image fitting and reconstruction capability, including hardware replay of those fitted checkpoints. The symmetry and restricted-entanglement diagnostics use the separate protocols stated in their respective sections; the exploratory training-dynamics material is addressed only as a provenance audit and is not retained as evidence. Held-out generalization and component necessity are evaluated under the resource-matched statistical protocol of the companion supplementary study (Section V-B). All settings report MSE and PSNR $= 20\log_{10}(1/\sqrt{\mathrm{MSE}})$; image settings additionally report SSIM. Implementation details are given in Appendix A.

IV. Results

A. Reconstruction Across Six Real Image Datasets

We trained a fresh HVK2D model per image (non-overlapping 8×8 patches, 16 patches/image, 80–100 optimization steps) on a representative sample of images from CIFAR-10 [27], MNIST [28], Fashion-MNIST [29], and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 [30] — the same six datasets used throughout this paper and its companion supplementary study. Table II reports the result: HVK2D reconstructs every dataset tested to a mean PSNR between 26.6 and 41.6 dB (SSIM 0.89–1.00), with the two medical-imaging datasets (BloodMNIST, PneumoniaMNIST) among the strongest results. This is a same-set, per-image capability result — it establishes that the architecture reconstructs real images of varied structure (natural photographs, handwritten digits, clothing, histopathology, blood-cell microscopy, and chest X-rays) well, not a held-out generalization claim; the companion supplementary study's held-out protocol is the appropriate evidence for the latter (Section V-B).

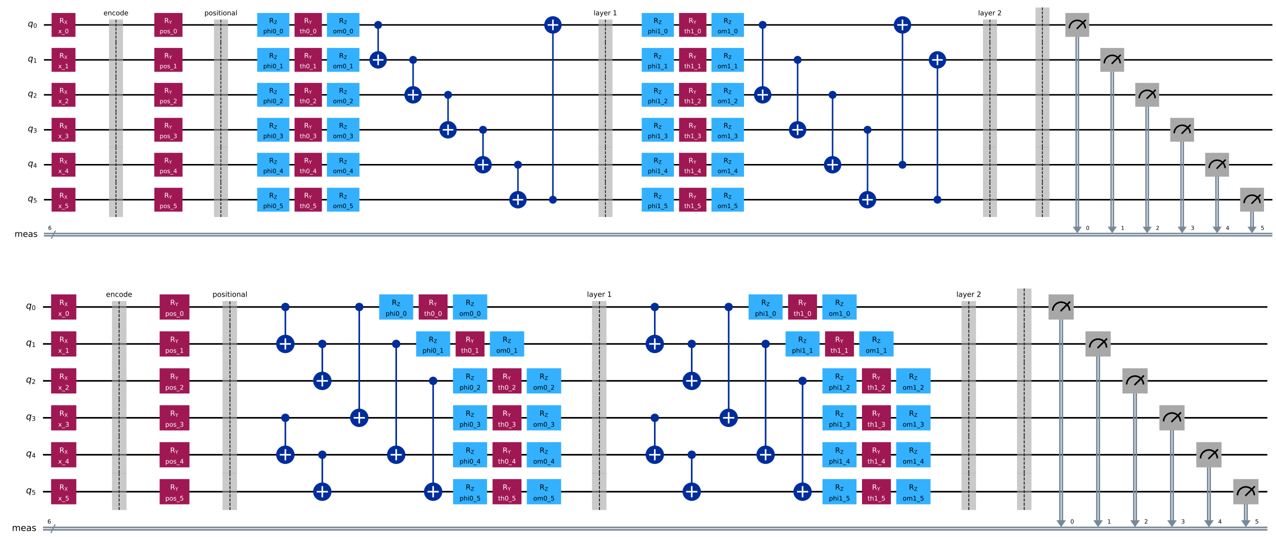

Fig. 1. The HVK1D (top) and HVK2D (bottom) variational ansätze, exactly as executed (encoding R_X rotations, positional R_Y rotations, then two entangling layers of R_Z-R_Y-R_Z single-qubit rotations with a CNOT ring for HVK1D or fixed grid-edge CNOTs for HVK2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section IV-C.

TABLE II
HVK2D SAME-SET RECONSTRUCTION ACROSS SIX REAL IMAGE DATASETS (PER-IMAGE TRAINED, 80–100 STEPS, MEAN ± STD OVER SAMPLED IMAGES).

Dataset	n images	PSNR (dB)	SSIM
CIFAR-10	3	37.52 ± 5.61	0.9978 ± 0.0016
MNIST	3	26.58 ± 2.01	0.9813 ± 0.0130
Fashion-MNIST	3	33.22 ± 1.34	0.9972 ± 0.0012
PathMNIST	3	34.97 ± 1.11	0.8907 ± 0.0597
BloodMNIST	2	33.07 ± 4.68	0.9707 ± 0.0273
PneumoniaMNIST	2	41.60 ± 0.98	0.9985 ± 0.0006

B. Scope and Adaptation Across Images

A model optimized on one image alone does not transfer zero-shot to a structurally different image: on *Monalisa*, zero-shot transfer to a second image reaches only 7.78 dB PSNR (SSIM $=$ 0.0196), while extending training to include the second image recovers 28.31 dB (SSIM $=$ 0.9862). This confirms that the per-image results above measure per-image optimization and reconstruction capability, not held-out generalization — consistent with how we scope every claim in this paper, and precisely the distinction the companion supplementary study's held-out protocol is designed to test (Section V-B).

C. A Real Hardware Reconstruction Pilot

Every result above is a noiseless-simulator result, a limitation we flagged rather than left implicit. We report a first step past it: a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), rather than the compiled-circuit feasibility probe of Appendix A alone. For five images — the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship*$\times 2$, *airplane*), each optimized directly on its own target image following the same per-image protocol as every result in this paper — we took an already-trained HVK1D/HVK2D checkpoint's exact learned parameters, rebuilt its measurement circuits gate-for-gate in Qiskit, verified the translation against the original PennyLane simulator to within numerical noise ($\leq 7.9 \times 10^{-4}$ maximum absolute error per observable across every patch; full validation numbers in the companion supplementary document), and then executed the same circuits on real hardware with 256 shots per measurement basis, a fixed per-basis shot budget rather than an adaptive shadow-tomography-style estimation scheme [25]. The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

Reconstructions from real, noisy hardware measurements retain recognizable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2); the four CIFAR-10 images average 29.56 ± 2.03 dB on hardware against 42.69 ± 3.84 dB in simulation (Figure 3, Table III). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. Every hardware reconstruction remains above the random-latent control range (11–15 dB in the companion supplementary study). The pilot used 176 circuits; the contemporaneous account-meter reading increased by 16 quantum-processor seconds, with the provenance limitation documented in the supplement.

Five images constitute a pilot, and we make no hardware-advantage claim. The narrower result is that the trained HVK observable channel remains visually and quantitatively informative in these device executions under the reported circuit and shot budget. Full methodology — exact gate sequences, per-basis measurement construction, job IDs, and pre-submission local-validation results — is documented in

Fig. 2. *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint's noiseless-simulator replay (center) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognizable.

TABLE III
REAL IBM HARDWARE RECONSTRUCTION PILOT (BACKEND `ibm_fez`, 256 SHOTS/BASIS, NO RETRAINING). PSNR COMPARES HARDWARE-DECODED AND SIMULATOR-DECODED RECONSTRUCTIONS AGAINST THE SAME GROUND-TRUTH IMAGE.

Image	Patches	Sim. PSNR (dB)	HW PSNR (dB)
Monalisa (HVK1D)	16	37.99	25.90
CIFAR cat (HVK2D)	16	44.85	31.52
CIFAR ship, hydrofoil (HVK2D)	16	36.56	26.44
CIFAR ship, sea boat (HVK2D)	16	42.57	31.20
CIFAR airplane (HVK2D)	16	46.79	29.10
CIFAR mean ± std (4 images)	–	42.69 ± 3.84	29.56 ± 2.03

the companion supplementary document.

D. Noise/Shot Trade-off: A Free Simulator Complement

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. Real IBM Quantum time is a scarce free-tier resource, so we separate what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (`FakeFez`, no quantum-processor time consumed), we replay the same already-trained checkpoints' circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap: moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots does not reliably recover it: increasing to 4096 shots per basis changes PSNR by only -0.05 dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a 0.96 dB loss — at this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality. Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture: a further 4.24 dB average gap (range 2.25–8.06 dB across images), attributable to device effects a single calibration snapshot does not fully capture — drift between calibration and execution, crosstalk, and correlated errors chief among them. The airplane image is the clear outlier (8.06 dB residual gap, roughly double any other image), which we report rather than exclude; a single-execution, five-image pilot cannot distinguish an image-specific effect from ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it: it bounds how much of the observed hardware gap is attributable to modeled device noise versus unmodeled effects, using zero additional quantum-processor time.

E. Provable D_4 Symmetry

A D_4-pooled HVK2D observable-grid map is equivariant to the eight symmetries of the square *by construction*: $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g\in D_4} P_g^{-1}\Phi(gx)$ averages the unpooled map over the group orbit, so $\Phi_{D_4}(hx) = P_h\Phi_{D_4}(x)$ for every $h \in D_4$ up to floating-point error. We verify this numerically over 7000 patch transforms of cached CIFAR images (Table IV, Figure 5): the pooled map's equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$ (floating-point noise), while the original unpooled positional map is not measurably more symmetric than a local/raw control (0.815 ± 0.129 vs. 0.837 ± 0.131) — confirming the unpooled map is only D_4-*motivated* by the patch grid's geometry, not empirically equivariant, and that only the explicit pooling construction delivers the property. This is currently a feature-grid-level construction and numerical diagnostic rather than a component of the trainable reconstruction pipeline; integrating it end-to-end (Section V-E) is concrete future work.

F. Entanglement Necessity Within the Tested Nonlocal-Structure Protocol

This diagnostic is deliberately constructed to reward pair-observable structure: the target is a fixed synthetic function of *simulated* patch features not concatenated into any candidate's own input block (Appendix A summarizes the leakage-audit

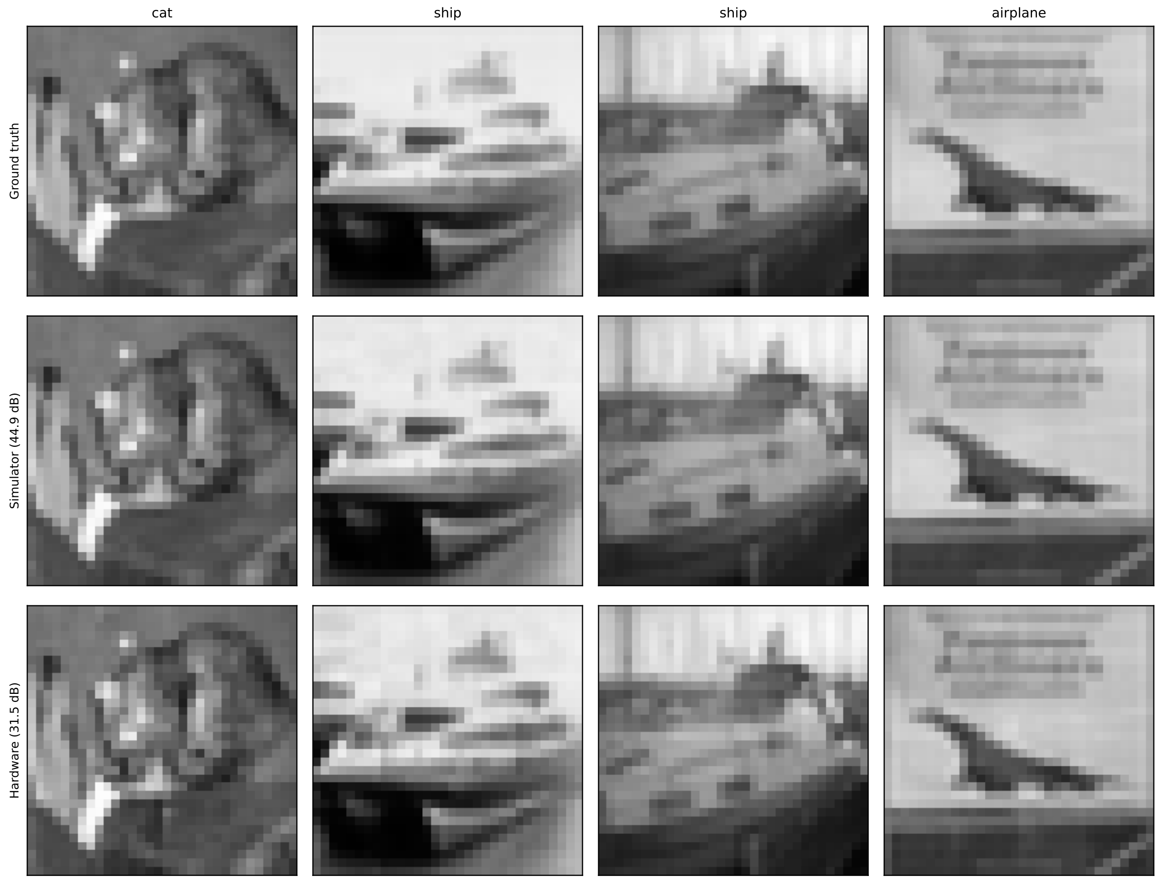

Fig. 3. Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained HVK2D checkpoints (middle row) and ground truth (top row).

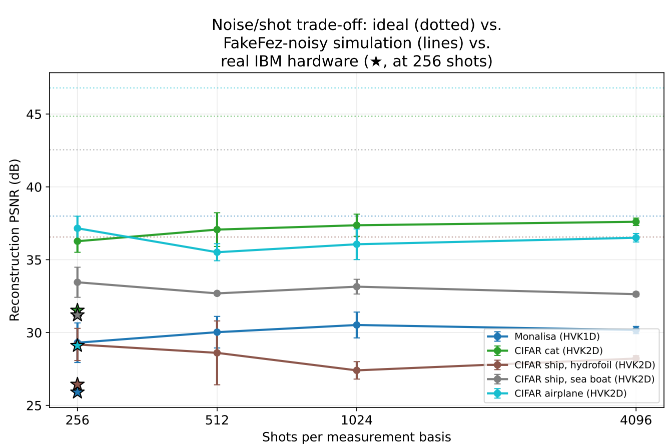

Fig. 4. Noise/shot trade-off from calibrated device-noise simulation (`FakeFez`, mean ± std over 3 repeated executions per shot count), against the noiseless-simulator ceiling (dotted, per image) and the real-hardware point from Table III (star, plotted at 256 shots for visual reference — the real job also used 256 shots). Shot count from 256 to 4096 changes PSNR by only -0.05 dB on average (range -0.96 to $+1.33$ dB per image) and does not close the gap to real hardware.

procedure that rules out circularity), and every control receives the same raw inputs as HVK2D. Table V and Figure 6 report the result across five seeds: HVK2D's entangling observable channel reaches mean $R^2 = 0.9735$ (32.77 dB PSNR), a 17.74 dB improvement over a no-entanglement control ($R^2 = 0.019$) and 18.52 dB over random-VQC latents ($R^2 = -0.173$); freezing the quantum circuit (features fixed, only readout trained) still reaches $R^2 = 0.853$, while a parameter-matched classical control ($R^2 = -0.030$) and a frozen-classical control ($R^2 = -0.903$) fail outright. This is evidence that an entangling pair-observable channel is *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target.

TABLE IV
D_4 EQUIVARIANCE ERROR FOR UNPOOLED AND POOLED OBSERVABLE-GRID MAPS (LOWER IS BETTER; POOLED MAP IS GROUP-AVERAGED OVER ALL EIGHT SQUARE SYMMETRIES).

Model	Mean error	Std. error
D_4-pooled HVK2D	9.57×10^{-17}	1.26×10^{-17}
No positional encoding	7.40×10^{-1}	1.52×10^{-1}
HVK2D positional	8.15×10^{-1}	1.29×10^{-1}
Local/raw	8.37×10^{-1}	1.31×10^{-1}

TABLE V
HVK2D RESTRICTED PAIR-CORRELATION DIAGNOSTIC ACROSS FIVE SEEDS. DELIBERATELY FAVORABLE TO PAIR-OBSERVABLE STRUCTURE; EVIDENCE FOR ENTANGLEMENT-NECESSITY UNDER A MATCHED LINEAR READOUT.

Model	Mean MSE	Mean PSNR (dB)	Mean R^2
HVK2D entangling observables	8.5969×10^{-4}	32.77	0.9735
Freeze quantum	4.7904×10^{-3}	27.34	0.8533
No entanglement	3.1461×10^{-2}	15.03	0.0191
Raw-linear classical	3.1654×10^{-2}	15.00	0.0133
Parameter-matched classical	3.3033×10^{-2}	14.82	-0.0297
Random VQC	3.7616×10^{-2}	14.25	-0.1732
Freeze classical	6.0973×10^{-2}	12.17	-0.9025

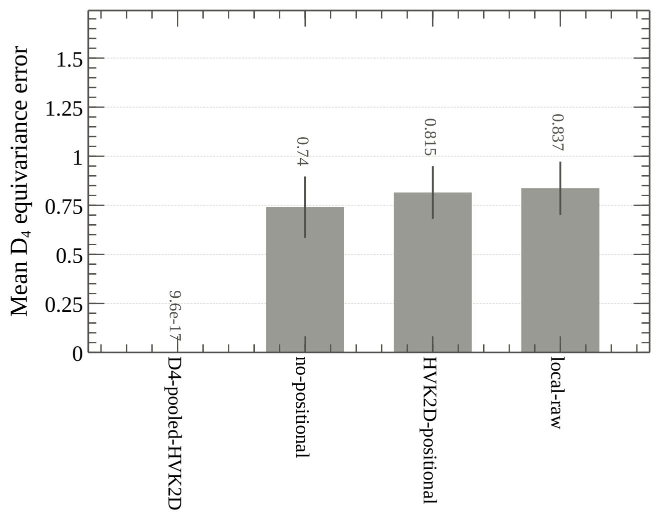

Fig. 5. D_4 equivariance diagnostic. The unpooled positional map is not equivariant; the group-pooled map is equivariant to numerical precision.

G. Training-Dynamics Reorganization: A Corrected Multi-Seed Result

An earlier version of this diagnostic was withdrawn after a provenance audit: the restricted-diagnostic trajectories previously shown were analytic interpolations produced for visualization rather than measurements from saved checkpoints, and the original multi-dataset traces were recorded during training, where HVK2D's training-mode forward pass deliberately adds Gaussian observable noise that confounds genuine parameter evolution with injected jitter. We report that finding rather than the original claim, and give the corrected replacement here rather than leave the diagnostic absent. The corrected runner evaluates a separate, noise-free forward pass in evaluation mode immediately after every optimizer update, and tracks the resulting global order parameter $M_z(t)$ and susceptibility $\mathcal{X}(t) = |M_z(t) - M_z(t - \Delta t)|$ across training, using the same median-plus-two-standard-deviation critical-epoch detection rule as before.

Table VI reports the result across all six datasets (two images $\times$ two seeds each, 24 runs total). Unlike the withdrawn claim, detection is not universal: MNIST detects a reorganization in all four runs, BloodMNIST and PneumoniaMNIST in three of four, and CIFAR-10, Fashion-MNIST, and PathMNIST in two of four, for an overall rate of $16/24$ (66.7%). We report this mixed result deliberately — a training-dynamics diagnostic that triggers on two-thirds of runs, with real dataset-to-dataset variation, is a more credible finding than the withdrawn claim's uniform 13-for-13 detection, and is the finding this corrected, noise-free protocol actually supports.

TABLE VI
CORRECTED, NOISE-FREE TRAINING-DYNAMICS DIAGNOSTIC ACROSS ALL SIX DATASETS (2 IMAGES $\times$ 2 SEEDS/DATASET).

Dataset	n	Detected	Mean t_c	Mean $\mathcal{X}_{\max}$
CIFAR-10	4	2/4	64.0	0.0036 ± 0.0009
MNIST	4	4/4	98.75	0.0023 ± 0.0012
Fashion-MNIST	4	2/4	45.0	0.0027 ± 0.0011
PathMNIST	4	2/4	54.5	0.0038 ± 0.0015
BloodMNIST	4	3/4	96.0	0.0033 ± 0.0012
PneumoniaMNIST	4	3/4	103.0	0.0036 ± 0.0008
Total	24	$16/24$	–	–

V. DISCUSSION

A. Task-Dependent Representational Scope

Whether a quantum feature map or kernel offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class, and dequantization results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched by classical algorithms once the right classical feature is identified [26]. The restricted diagnostic of Section IV-F is constructed so that the relevant structure is a genuine distant-element product raw or local features cannot represent under a fixed linear readout, and that is precisely the regime in which HVK's entangling channel separates. Natural image patches, at the resolution and patch size used in ordinary reconstruction, are a different regime — low-entanglement, locally structured data where a linear or shallow classical readout is already close to sufficient. The companion supplementary study's held-out comparisons (Section V-B) characterize exactly this second regime; together, the results delineate where the tested entangling representation does and does not yield a measurable benefit.

B. Relationship to the Component-Wise Ablation Study

The results above establish what HVK is and several of its provable or verified properties. A separate, equally important question — whether its feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol — is answered in the companion document,

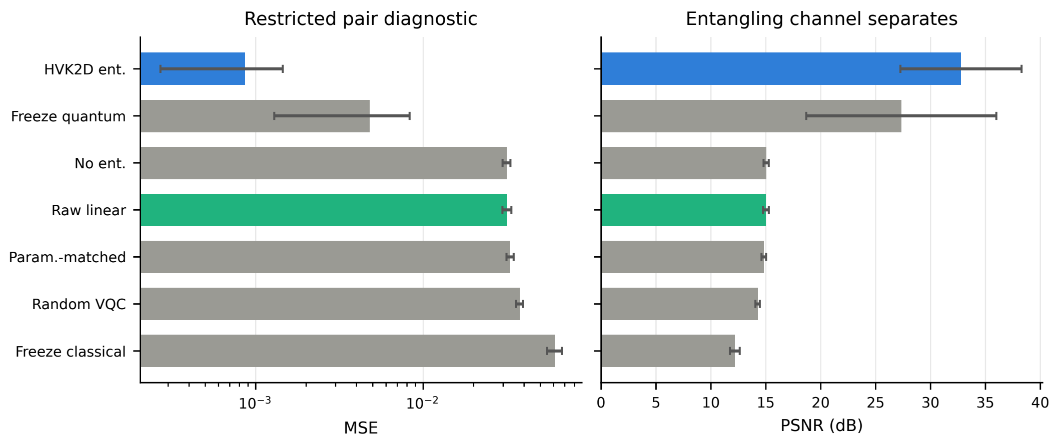

Fig. 6. Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The entangling observable feature map dominates every control on this deliberately favorable task.

Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility. On held-out CIFAR-10, local-observable and raw-linear controls reach 18.80 ± 1.42 dB against HVK2D's 18.12 ± 1.54 dB; after aggregating held-out images within each split, the HVK2D-minus-control difference is -0.68 dB (bootstrap 95% CI $[-0.91, -0.46]$ dB; Wilcoxon $p = 0.0625$, $n = 5$ seeds), with the same numerical direction on five further real datasets. Thus the tested HVK map is informative but is not shown to outperform the simpler maps on ordinary held-out reconstruction. We report this as a rigorous complement: it delineates the tested boundary of the positive entanglement and symmetry results and records the provenance checks that prevented exploratory artifacts from becoming claims.

C. Topology Alignment as an Inductive Bias

Aligning the latent interaction graph with image-patch adjacency (HVK2D vs. HVK1D) is a plausible architectural inductive bias: on *Monalisa*, 2D HVK exceeds its 1D counterpart by 5.98 dB (40.70 vs. 34.72 dB). The companion supplementary study tests this hypothesis more directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives an HVK1D-minus-HVK2D mean PSNR difference of 0.16 dB; after averaging images within seed, the bootstrap 95% CI is $[-0.38, 0.46]$ dB and the seed-level Wilcoxon test gives $p = 0.50$. HVK2D completes the matched 90-step runs in roughly half the wall-clock time. A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences (at most 0.07 dB in mean PSNR). At the tested scale, topology therefore appears to be primarily a resource-cost choice rather than a reliable accuracy advantage; these tests do not establish formal equivalence.

D. Architectural Differentiation Matrix

Table VII contrasts HVK's structural properties against reference frameworks. Its distinguishing features — a Pauli-correlator latent space, 2D Fourier positional bias with enforced D_4 pooling, and a Hamiltonian energy diagnostic — are documented and scoped in the preceding sections.

E. Validated Scope and Next Steps

- Every reconstruction result in this paper (Sections IV-A–IV-B) is per-image trained, not held-out generalization; the companion supplementary study's held-out protocol is the appropriate evidence for a generalization claim, and its finding that local/raw maps match or exceed the tested HVK map under the shared readout should be read alongside the architecture-level results here.
- The D_4-pooled map (Section IV-E) is currently a feature-grid-level construction and numerical diagnostic, not yet integrated into the trainable end-to-end reconstruction pipeline; doing so is concrete future work (Section VI).
- The entanglement-necessity result (Section IV-F) is scoped to a task deliberately constructed to require nonlocal structure; it does not by itself establish an advantage on ordinary natural-image reconstruction (see Section V-A).
- The hardware pilot (Section IV-C) is a five-image, single-trial, unmitigated pilot, not a statistically resolved hardware benchmark.
- The original exploratory training-dynamics traces were withdrawn after a provenance audit (Section IV-G); the corrected, noise-free multi-seed replacement is a small-sample diagnostic (2 images $\times$ 2 seeds/dataset, 24 runs across six datasets) with mixed, not universal, detection ($16/24$), and should be read at that scope.
- The matched real-circuit topology comparison uses only 3 seeds and 3 held-out images per topology; the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation but do not establish formal equivalence of the topologies.

VI. Conclusion

We presented the Hamiltonian Vision Kernel, a physics-informed hybrid quantum–classical image autoencoder distinguished by a Pauli-correlator latent space, an enforced

TABLE VII
ARCHITECTURAL DIFFERENTIATION AGAINST REPRESENTATIVE BASELINE DESIGNS EVALUATED OR DISCUSSED IN THIS WORK. ENTRIES SUMMARIZE THOSE REFERENCE DESIGNS, NOT EVERY MODEL IN EACH BROAD ARCHITECTURE FAMILY.

Axis	Conv. AE baseline	GAN baseline	Register-based QAE	HVK (ours)
Latent representation	Continuous $\mathbb{R}^d$	Continuous $\mathbb{R}^d$	Quantum register	Pauli-correlator vector
Spatial bias	Convolutional	Convolutional	Encoding-dependent	2D Fourier encoding
Auxiliary objective	Bottleneck/reconstruction	Adversarial loss	Fidelity/reconstruction	Hamiltonian diagnostic
Enforced group symmetry	Not in baseline	Not in baseline	Not in reference design	D_4 observable pooling
Evaluated role	Reconstruction	Reconstruction	Compression	Reconstruction + diagnostics

and numerically verified D_4-equivariant observable-pooling extension, and a learnable graph-Hamiltonian energy diagnostic. Independently of reconstruction quality, we established two structural results at the tested scope: a provable D_4 symmetry construction (equivariance error 9.57×10^{-17}) and a restricted entanglement-sensitive result on a task requiring nonlocal structure, where pair observables are necessary within the tested fixed-readout feature class ($R^2 = 0.9735$ vs. $R^2 \leq 0.02$ for non-entangling controls). We further reported a real IBM Quantum hardware reconstruction pilot: replaying trained checkpoints without retraining yielded 25.90–31.52 dB PSNR from hardware measurements. This pilot establishes feasibility at its five-image, single-trial scope rather than a general hardware-performance claim.

A companion supplementary study reports a separate, rigorous question: does the tested HVK feature map improve reconstruction on held-out real images under resource-matched controls? It does not at the evaluated scale; local/raw maps match or exceed it under the same fitted readout. This is not in tension with the structural results here. Instead, it separates them from an ordinary-reconstruction advantage: entangling observables separate only on the tested nonlocal target, and group pooling guarantees equivariance without establishing an accuracy gain. We consider this carefully measured boundary to be a constructive contribution alongside the architecture itself.

Future work should integrate the D_4-pooled map into an end-to-end trainable pipeline (currently a feature-grid-level construction, Section V-E); scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials; search for further real-world tasks with genuine nonlocal structure where HVK's restricted entanglement-sensitive separation translates into a reconstruction advantage; and extend the resource-matched, leakage-audited evaluation protocol used in the companion supplementary study to other hybrid quantum vision architectures.

DATA AND CODE AVAILABILITY

The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps the three numerical headline claims to their retained source artifacts, and automated tests verify those mappings. Dataset provenance and access routes are cited in the manuscript and detailed in the supplementary artifact map. IBM Quantum job identifiers and reconstructed outputs are retained; as disclosed in the supplement, the raw account-usage service response was not serialized, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

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APPENDIX

For a one-dimensional chain of n_{bonds} bonds indexed by i and sites $(i, i+1)$, the anisotropic Heisenberg Hamiltonian is

$$H_{\text{Heis}} = \sum_{i=0}^{n_{\text{bonds}}-1} \Big(J_x^{(i)} X_i X_{i+1} + J_y^{(i)} Y_i Y_{i+1} + J_z^{(i)} Z_i Z_{i+1} \Big), \tag{2}$$

where X, Y, Z are Pauli operators and $J_\alpha^{(i)}$ are bond-resolved couplings. HVK1D fixes six sites and treats the 15 coefficients on its five bonds as learnable parameters initialized independently from a zero-mean Gaussian with standard deviation 0.1. Given patch n's measured observables, its energy diagnostic is

$$\mathcal{E}_n(\theta, J) = \sum_{i=0}^{4} \sum_{\alpha \in \{x,y,z\}} J_\alpha^{(i)} \langle \sigma_\alpha^{(i)} \sigma_\alpha^{(i+1)} \rangle_n, \tag{3}$$

HVK2D instead uses the seven edges $\mathcal{G}_{2D}$ of its 2×3 circuit graph and the implemented grid-Ising diagnostic $\mathcal{E}_n^{2D} = \sum_{(u,v) \in \mathcal{G}_{2D}} j_{uv} \langle Z_u Z_v \rangle_n$. Thus "Hamiltonian diagnostic" refers to the full $XX/YY/ZZ$ chain energy for HVK1D and the learned ZZ graph energy for HVK2D. In either case, the patch-averaged term is $\mathcal{L}_{\text{energy}} = \frac{1}{N} \sum_n \mathcal{E}_n$, and the total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda \, \mathcal{L}_{\text{energy}}, \quad \mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{n=1}^{N} \big\| D(o_n, e_n) - P_n \big\|_F^2, \tag{4}$$

with D the classical decoder and P_n the ground-truth patch.

Per-patch coordinates (r_i, c_j) are mapped to Fourier features $e_{4k} = \sin(\omega_k \pi r_i)$, $e_{4k+1} = \cos(\omega_k \pi r_i)$, $e_{4k+2} = \sin(\omega_k \pi c_j)$, and $e_{4k+3} = \cos(\omega_k \pi c_j)$ [15]. HVK1D uses $k \in \{0, 1\}$ (8-D), whereas the native-resolution HVK2D implementation uses one frequency (4-D); a learned linear map projects either vector to six R_Y angles. Both six-qubit circuits first apply $U_{\text{enc}}(x_n) = \bigotimes_{i=0}^{5} R_X(x_n^{(i)})$ and $U_{\text{pos}} = \bigotimes_{i=0}^{5} R_Y(\theta_{\text{position}}^{(i)})$. Each of two HVK1D layers then applies per-qubit Rot gates followed by the strongly-entangling CNOT ring. Each HVK2D layer applies CNOTs over the four horizontal and three vertical

TABLE VIII
CIRCUIT-RESOURCE SUMMARY FOR HVK PATCH CIRCUITS MAPPED TO AN IBM HERON-CLASS BASIS.

Variant	Qubits	Depth	R_Y	R_Z	CNOT
HVK1D	6	18	11	6	10
HVK2D	6	18	13	6	14

grid edges followed by per-qubit Rot gates, matching the executed source and hardware-replay circuits.

HVK1D returns local Z/X measurements and the three nearest-neighbor pair sectors $ZZ/XX/YY$ (27 values); HVK2D returns local Z/X measurements and grid-edge ZZ correlations (19 values). Positional rotations and fixed graph connectivity distinguish circuit sites, so neither unpooled map is asserted to be permutation- or D_4-equivariant. The D_4 action used in Section IV-E is instead the explicit permutation action on the 4×4 patch grid, and equivariance follows only after the stated Reynolds average over its eight transformations.

Adam optimizer, $\eta = 0.003$ (1D) / $\eta = 0.004$ (2D); energy regularization $\lambda = 0.01$ unless explicitly ablated. The Monalisa/HVK1D MPS extraction unrolls each 64×64 patch to 2^{12} amplitudes and produces the 46-D feature vector detailed above. The native 32×32 HVK2D runs analogously map each 8×8 patch to 2^6 amplitudes and a 22-D feature vector. Both use sequential SVD compression with $\chi = 4$, probabilities normalized with $\varepsilon = 10^{-8}$ in the entropy calculation, feature standardization over the 16 patches, and a learned projection to six circuit inputs.

The entanglement-necessity diagnostic of Section IV-F was independently audited for target–feature leakage: the target is a fixed synthetic function of simulated patch features not concatenated into any candidate model's own input block, so no model can achieve fit quality by copying precomputed target-construction quantities from its own features. This audit was itself the product of catching an earlier, circular benchmark of our own (an entangling observable channel reaching a spurious $R^2 \approx 1$ by copying, not learning, distant-patch product terms that had been concatenated onto its own input); the full post-mortem and the resulting general-purpose leakage-detection rule are given in the companion supplementary document.

Table VIII and Figure 7 report circuit resources for the HVK1D and HVK2D patch circuits compiled to an IBM Heron-class basis (patch index 0). Both variants use six measured qubits and depth-18 circuits after transpilation; HVK2D requires more CNOT gates because it encodes additional grid-adjacency links. This is a compiled-circuit feasibility check across the broader variant sweep, complementary to the finite-shot reconstruction pilot of Section IV-C: this appendix draws no image-quality conclusion by itself, while Section IV-C reports the actual hardware-measured reconstruction numbers, with full methodology in the companion supplementary document.

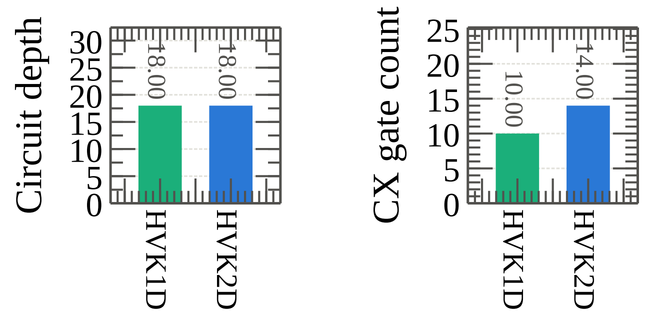

Fig. 7. IBM Cloud circuit-resource summary for the HVK1D and HVK2D hardware probes.

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